Midterm Review

Xiao Zhang

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Part I. Basic Statistical Concepts and Data Collection

- Define and distinguish between **statistics**, **probability**, **population**, **sample**, **parameter**, and **statistic**.
- Distinguish among common **sampling methods**:
  - Simple random sampling
  - Stratified sampling
  - Cluster sampling
  - Systematic sampling
  - Convenience sampling
- Identify different **types of variables**:
  - Quantitative (discrete vs. continuous)
  - Qualitative / Categorical (binary, nominal, ordinal)
- Understand principles of **ethical data collection and experimentation**, including informed consent, random assignment, and avoiding harm.
- Recognize common **sources of bias**: selection bias, response bias, nonresponse bias, measurement bias, and undercoverage.

Part II. Descriptive Statistics and Data Visualization

- Interpret common **data displays**: histograms, boxplots, scatterplots, bar charts, and pie charts.
- Calculate and interpret **measures of center**: mean, median, and mode.
- Use sigma notation and correctly apply the sample mean formula:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

- Calculate and interpret **measures of spread**: range, interquartile range (IQR), variance, and standard deviation.

- Use and interpret the formulas:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

- Understand distribution **skewness** and the relationship between mean and median.
- Understand the effect of **linear transformations** on measures of center and spread.

Part III. Probability Concepts and Counting Methods

Basic Probability Rules

- Understand basic probability terminology: random experiment, sample space, event, probability, and complement.
- Distinguish between **independent** and **mutually exclusive** events.
- Apply the addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

- Apply the multiplication rule:

$$P(A \cap B) = P(A)\,P(B \mid A).$$

- Calculate conditional probabilities:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

- Apply the Law of Total Probability:
  If $\{B_1, \dots, B_k\}$ is a partition of the sample space with $P(B_i) > 0$, then

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i)\,P(B_i).$$

- Use contingency tables to compute marginal, joint, and conditional probabilities.
- Determine whether two events are independent using probability calculations.

Counting Techniques

- Recognize when counting methods are needed to compute probabilities.
- Distinguish between **permutations** (order matters) and **combinations** (order does not matter).
- Apply permutation formulas:

$$_nP_r = \frac{n!}{(n-r)!}.$$

- Apply combination formulas:

$$_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

- Identify real-world scenarios where permutations or combinations are appropriate.

Part IV. Discrete Random Variables

- Understand the definition of a **random variable** and how to define one in context.
- Distinguish between discrete and continuous random variables.
- Recognize and apply the **binomial distribution**:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad E(X) = np, \qquad SD(X) = \sqrt{np(1-p)}.$$

- Recognize and apply the **Poisson distribution**:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad E(X) = \lambda, \qquad Var(X) = \lambda.$$

- Choose an appropriate discrete distribution for a given scenario.
- Apply linear transformations of random variables:

$$E(aX + b) = aE(X) + b, \qquad Var(aX + b) = a^2 Var(X).$$